

Exp. No. : 1

Exp. Name: Effect of Additive White Gaussian Noise (AWGN) on Signals and Channel Capacity.

Aim: To study the effect of Additive White Gaussian Noise (AWGN) on a sinusoidal signal and to understand how noise limits the maximum data transmission rate using Shannon's Channel Capacity theorem.

Objectives:

By the end of this experiment, students will be able to:

1. Generate a continuous-time signal in MATLAB.
2. Add AWGN to a signal at different SNR levels.
3. Observe how signal quality degrades as noise increases.
4. Understand the concept of Signal-to-Noise Ratio (SNR).
5. Apply Shannon–Hartley theorem to calculate channel capacity.

Overview:

In real communication systems, transmitted signals are always affected by noise. The most common noise model is **Additive White Gaussian Noise (AWGN)**.

- **Additive** → Noise is added to the signal.
- **White** → Noise power is equal across all frequencies.
- **Gaussian** → Noise amplitude follows a normal distribution.

The received signal is:

$$r(t) = s(t) + n(t)$$

Signal quality is measured using:

$$SNR = \frac{P_{signal}}{P_{noise}}$$

In digital communication, noise limits the maximum achievable data rate. According to **Shannon's theorem**:

$$C = B \log_2(1 + SNR)$$

Where:

- C = Channel capacity (bits/sec)
- B= Bandwidth (Hz)
- SNR = Signal-to-noise ratio (linear)

Part A:

Part A Procedure: AWGN Effect on Signal

1. Open MATLAB – new script.
2. Define signal parameters:
 - Sampling frequency = 1000 Hz
 - Signal frequency = 5 Hz
 - Duration = 1 second [t = 0:1/fs:1]
3. Generate a clean sinusoidal signal [**clean signal = sin(2πft)**]
4. Select SNR values: **20 dB, 10 dB, 5 dB, 0 dB**.
5. Use MATLAB **awgn()** function to add noise.
6. Plot clean vs noisy signal for each SNR.
7. Observe waveform distortion as SNR decreases.

Part A MATLAB Code:

```
clear all; close all; clc;

% Parameters
fs = 1000;           % Sampling frequency (Hz)
t = 0:1/fs:1;        % Time vector (1 second)
f_signal = 5;         % Signal frequency (Hz)

% Generate clean signal
clean_signal = sin(2*pi*f_signal*t);

% Add different levels of AWGN
SNR_dB = [20, 10, 5, 0]; % Signal-to-Noise Ratios in dB
noisy_signals = zeros(length(SNR_dB), length(clean_signal));

for i = 1:length(SNR_dB)
    noisy_signals(i,:) = awgn(clean_signal, SNR_dB(i), 'measured');
end

% Plot results
figure('Name', 'AWGN Effects on Sinusoidal Signal');
for i = 1:length(SNR_dB)
    subplot(2,2,i);
    plot(t, clean_signal, 'b', 'Linewidth', 1.5); hold on;
    plot(t, noisy_signals(i,:), 'r', 'Linewidth', 0.5);
    title(['SNR = ', num2str(SNR_dB(i)), ' dB']);
    xlabel('Time (s)'); ylabel('Amplitude');
    legend('Clean Signal', 'Noisy Signal');
    grid on;
end
```

Results (A):

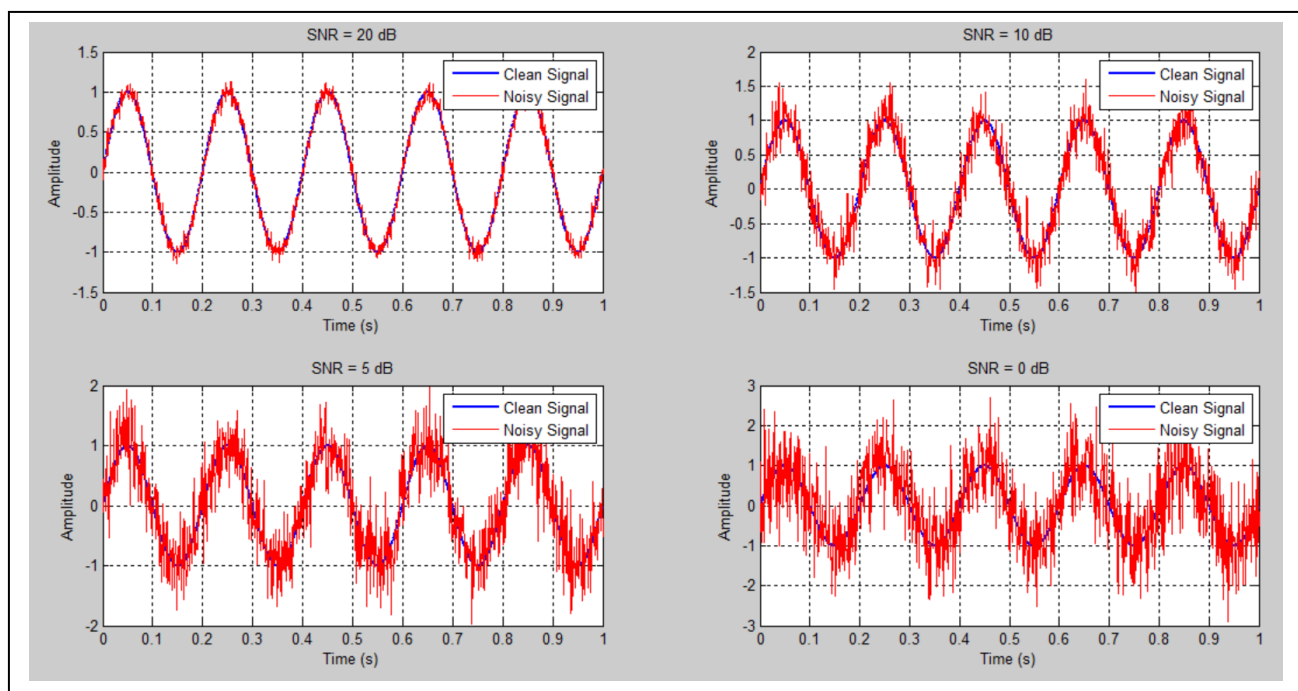


Fig (1.1)

Part B:

Part B Procedure: Channel Capacity

1. Define SNR range from **−10 dB to 30 dB**.
2. Convert SNR from dB to linear scale.
3. Set bandwidth **B= 1 MHz**.
4. Use Shannon formula to compute channel capacity.
5. Plot **Capacity vs SNR**.
6. Mark capacity at **0, 10, 20, 30 dB**.

Part B MATLAB Code:

```
% Parameters
SNR_dB = -10:0.5:30;          % SNR range in dB
SNR_linear = 10.^(SNR_dB/10); % Convert to linear scale
B = 1e6;                      % Bandwidth (1 MHz)

% Calculate channel capacity
C = B * log2(1 + SNR_linear); % Shannon capacity in bits/sec

% Plot
figure('Name', 'Shannon Channel Capacity');
plot(SNR_dB, C/1e6, 'Linewidth', 2);
grid on;
xlabel('SNR (dB)'); ylabel('Channel Capacity (Mbps)');
title(['Shannon Channel Capacity (Bandwidth = ', num2str(B/1e6), ' MHz)']);

% Add markers for specific SNR values
hold on;
SNR_markers = [0, 10, 20, 30];
C_markers = B * log2(1 + 10.^(SNR_markers/10));
plot(SNR_markers, C_markers/1e6, 'ro', 'MarkerSize', 10, 'Linewidth', 2);
for i = 1:length(SNR_markers)
    text(SNR_markers(i), C_markers(i)/1e6, ...
        sprintf(' %.2f Mbps', C_markers(i)/1e6));
end
```

Results (B):

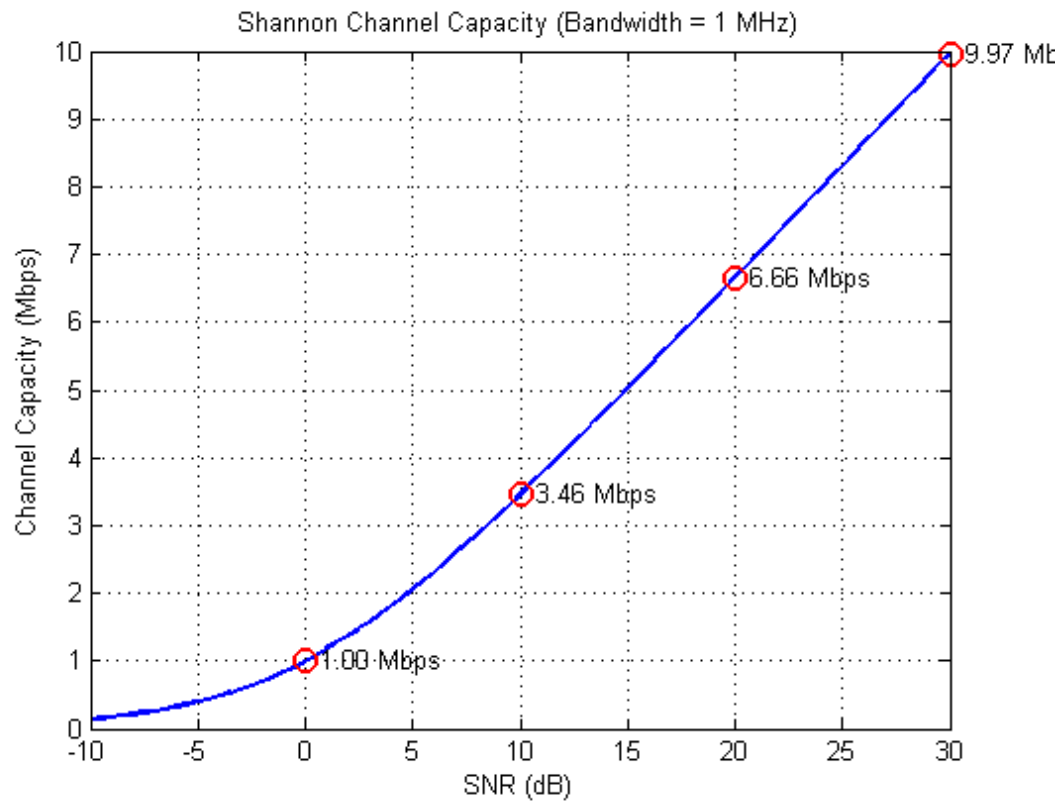


Fig (1.2)

Discussion:

1. What happens to the signal waveform as SNR decreases?
2. What does “white” mean in AWGN?
3. At which SNR does the signal become difficult to recognize?
4. What happens to capacity(**Mbps**) if bandwidth doubles? Draw the Mbps to SNR relationship when it doubled.